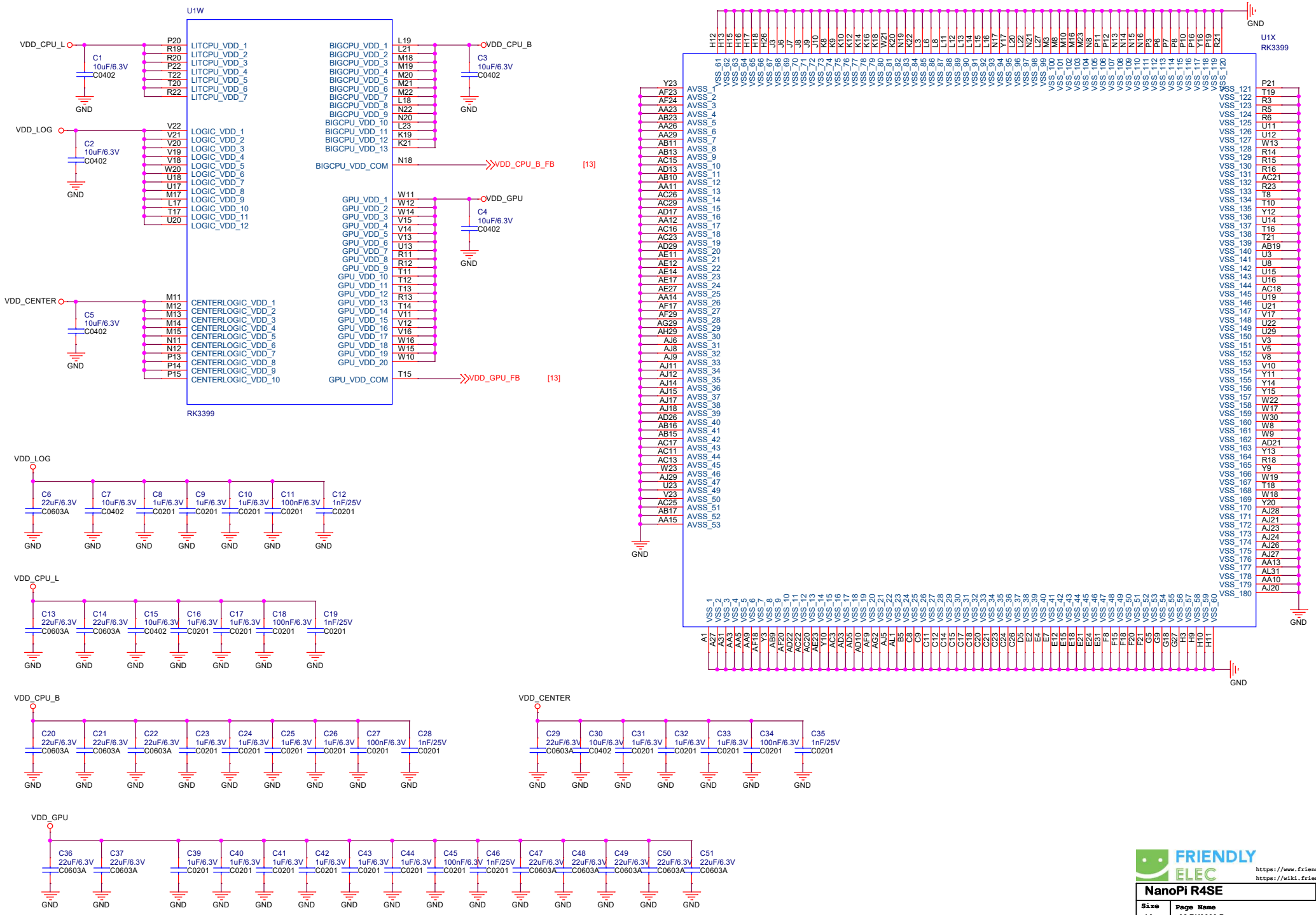
NanoPi R4SE

FRIENDLY ELEC

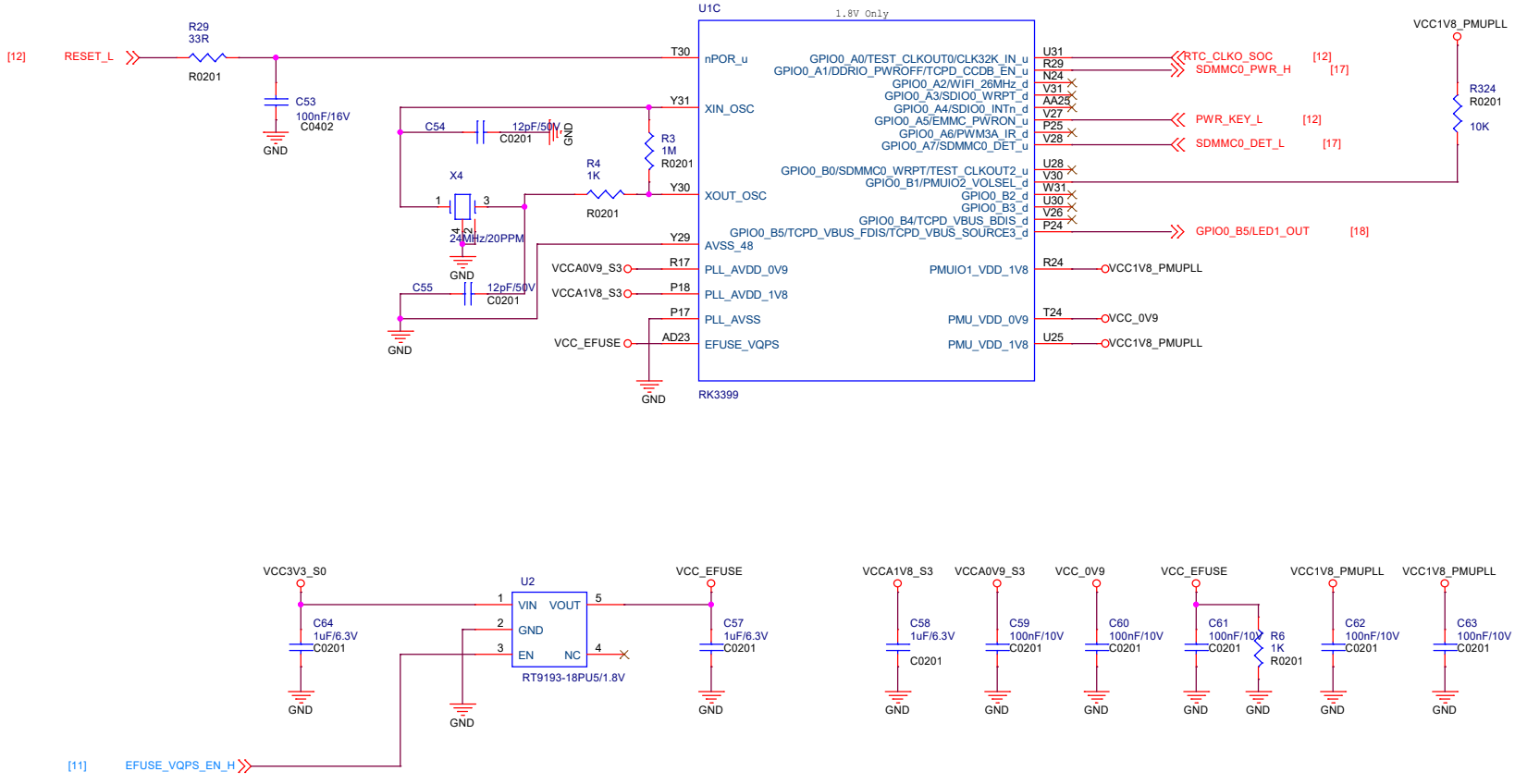
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE		Rev 2205
Size A3	Page Name 01.Title	
Date: Wednesday, May 25, 2022		Sheet: 1 / 21



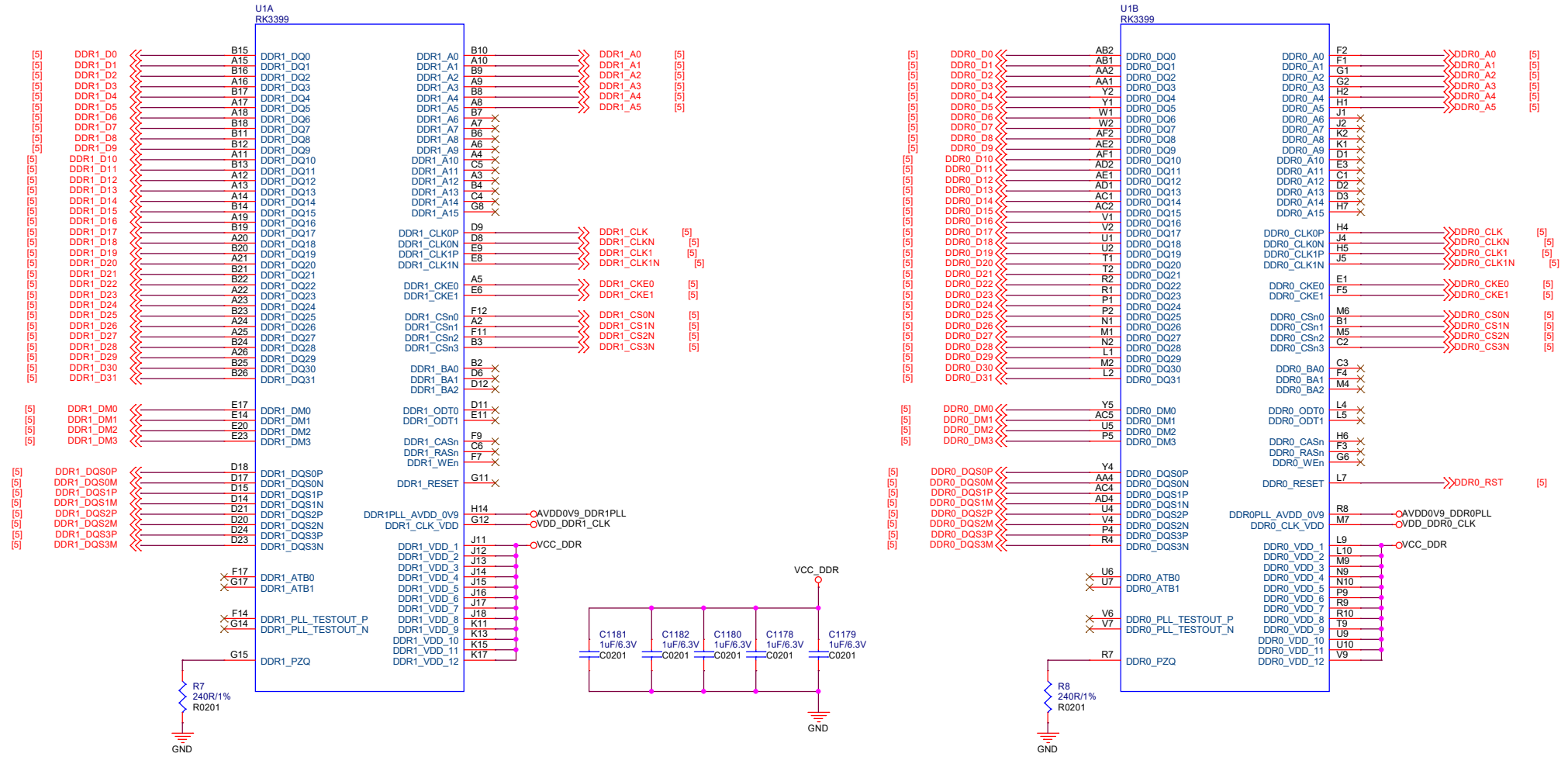
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 02.RK3399 Power
Date: Wednesday, May 25, 2022	Sheet: 2 / 21

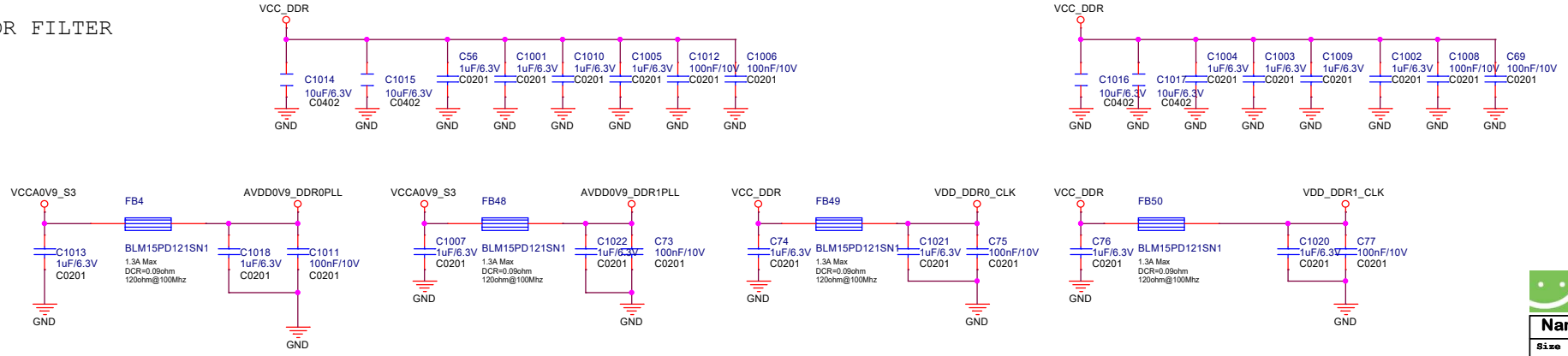


FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 03.RK3399 PMU Controler
Date: Wednesday, May 25, 2022	Sheet: 3 / 21

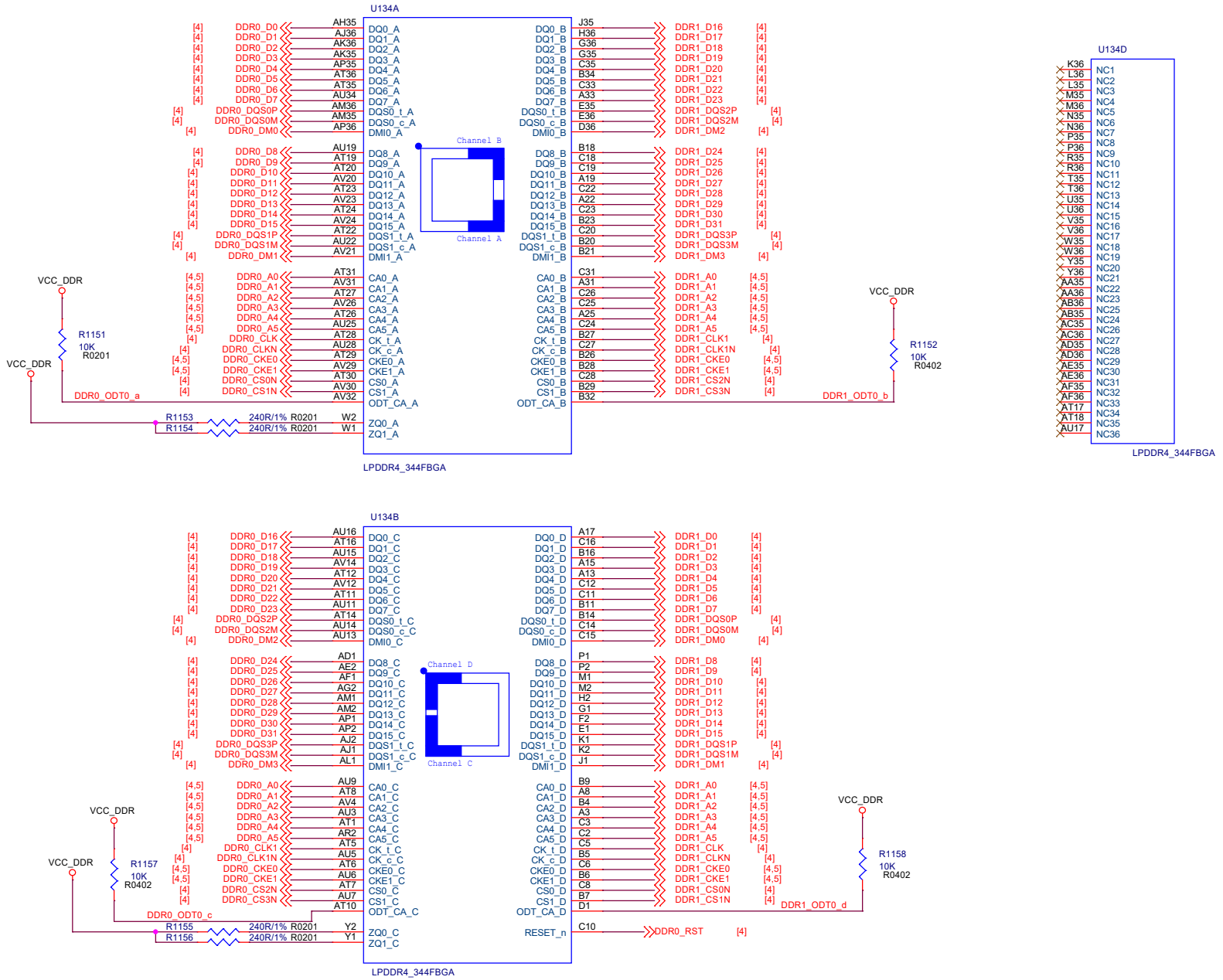


DDR FILTER



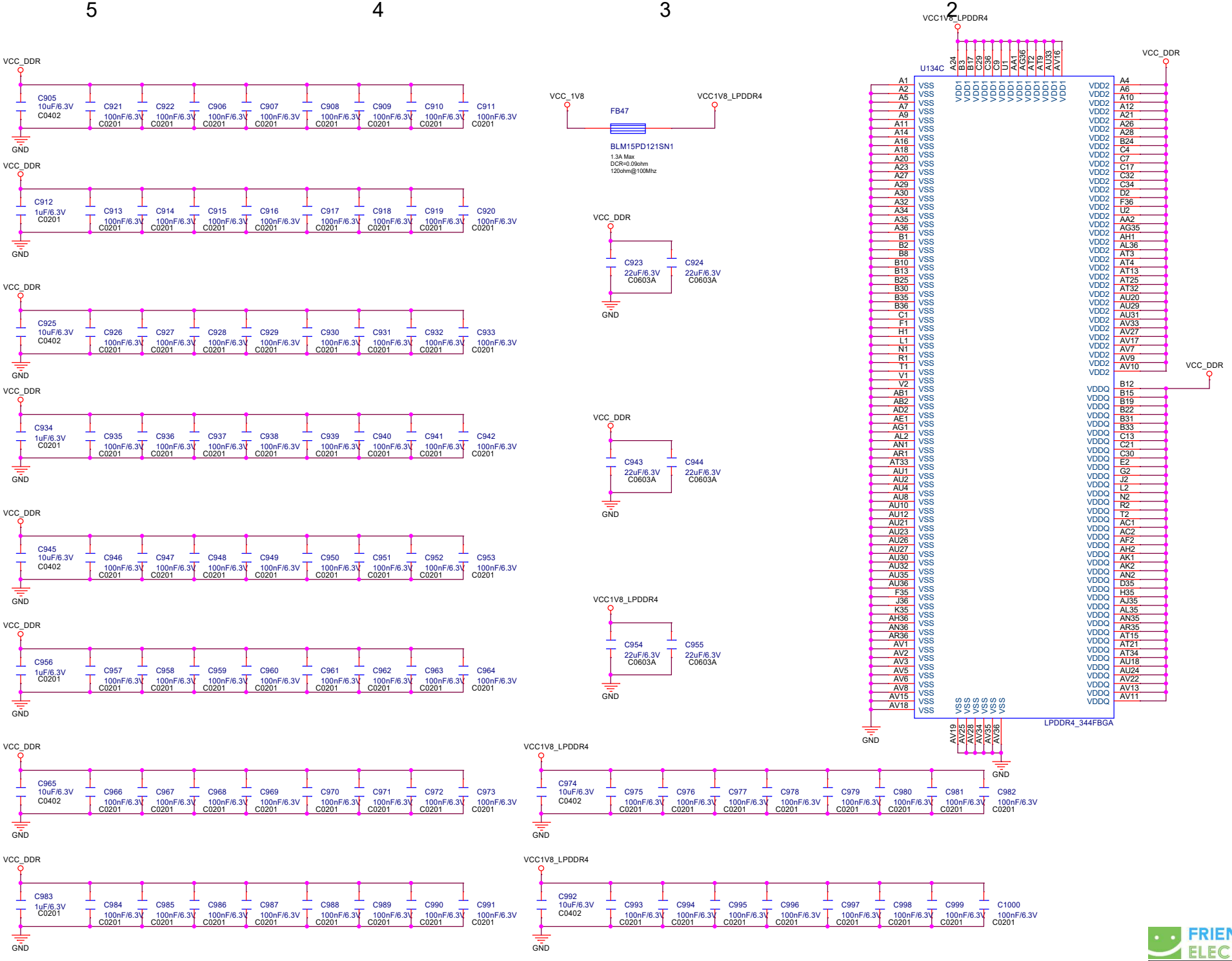
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE		Rev 2205
Size A3	Page Name 04.RK3399 DDR Controler	
Date: Wednesday, May 25, 2022		Sheet: 4 / 21



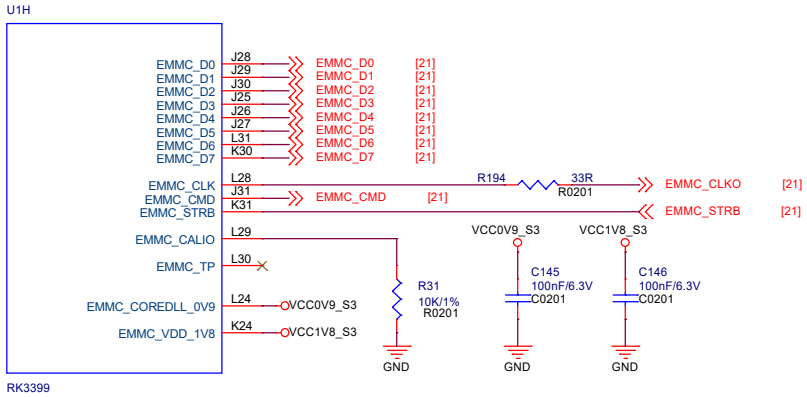
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

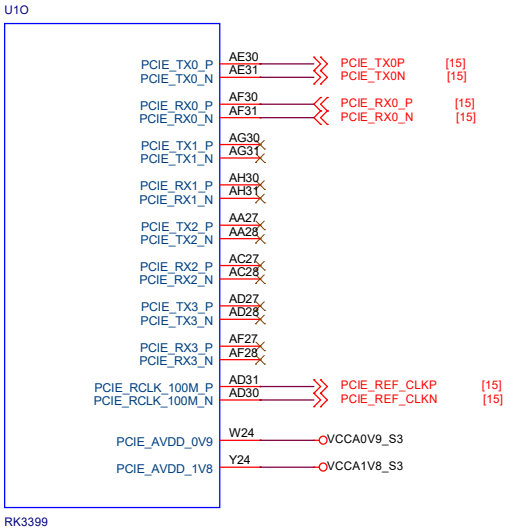
NanoPi R4SE	Rev 2205
Size A3	Page Name 05.LPDDR4_A
Date: Wednesday, May 25, 2022	Sheet: 5 / 21

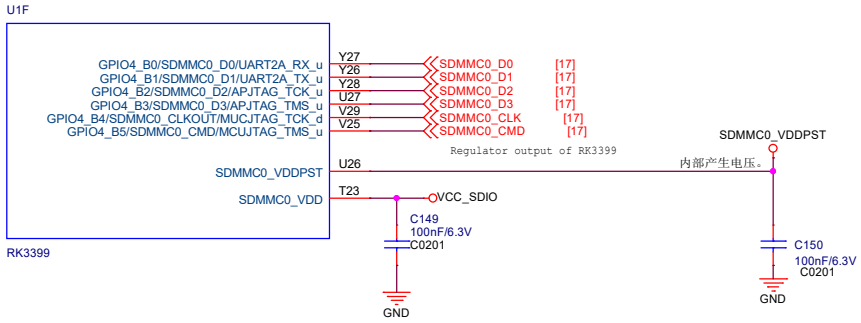


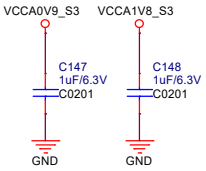
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

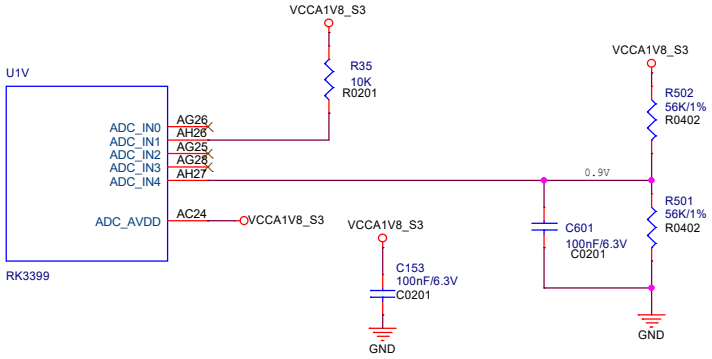
NanoPi R4SE	Rev 2205
Size A3	Page Name 06.LPDDR4_B
Date: Wednesday, May 25, 2022	Sheet: 6 / 21





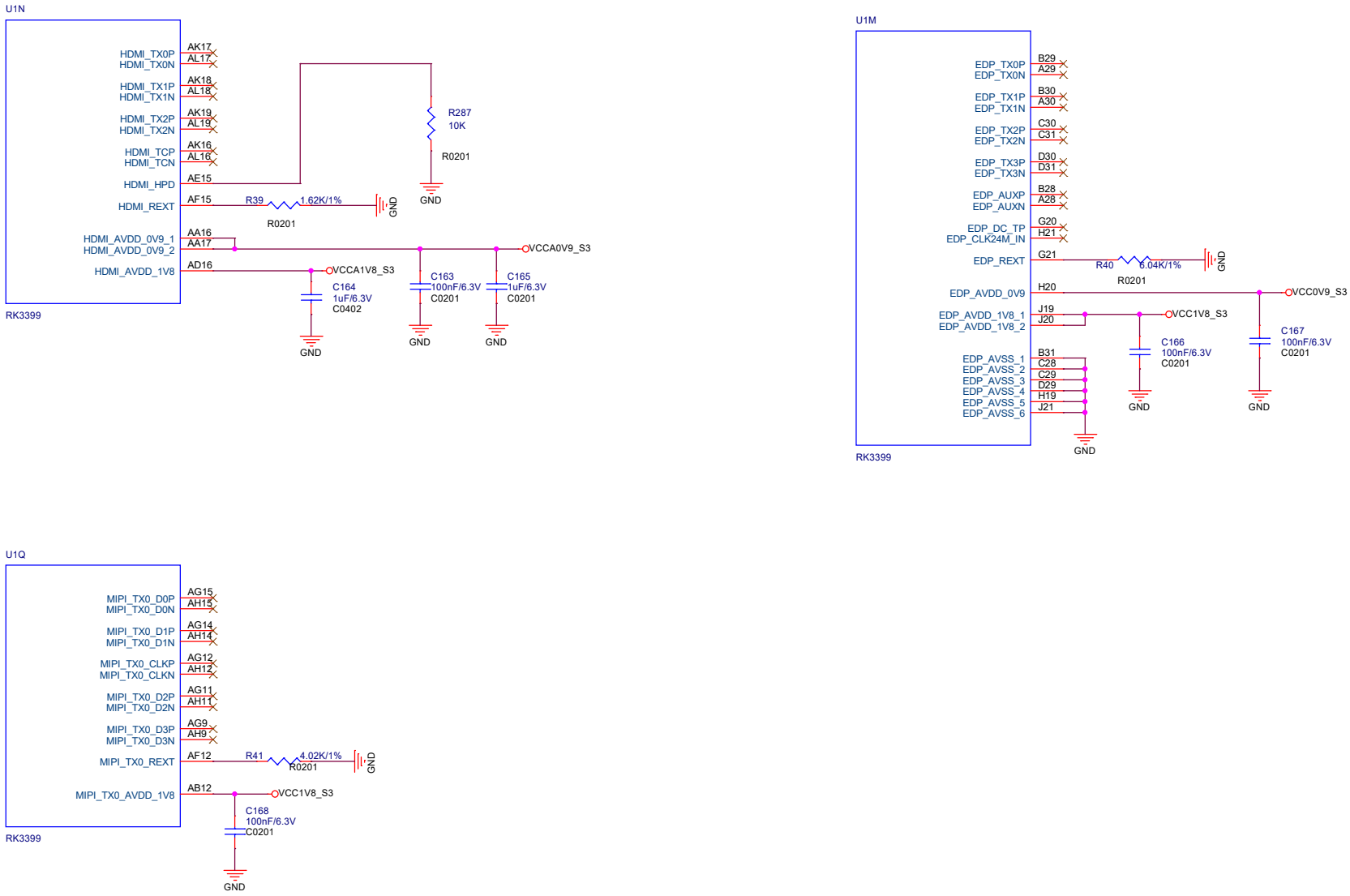






FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

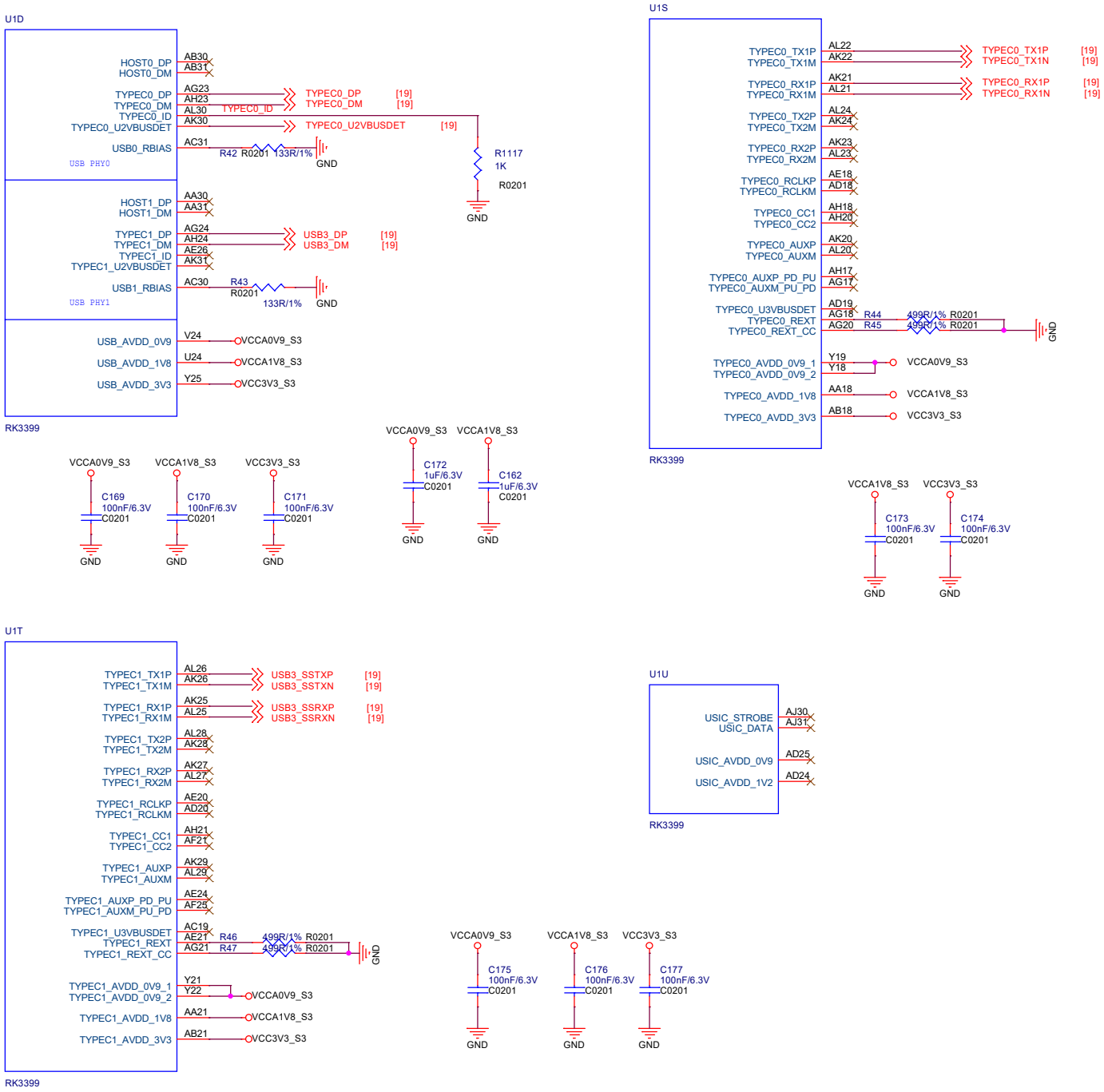
NanoPi R4SE	Rev 2205
Size A3	Page Name 07.RK3399 MMC/ADC/PCIe
Date: Wednesday, May 25, 2022	Sheet: 7 / 21



FRIENDLY ELEC

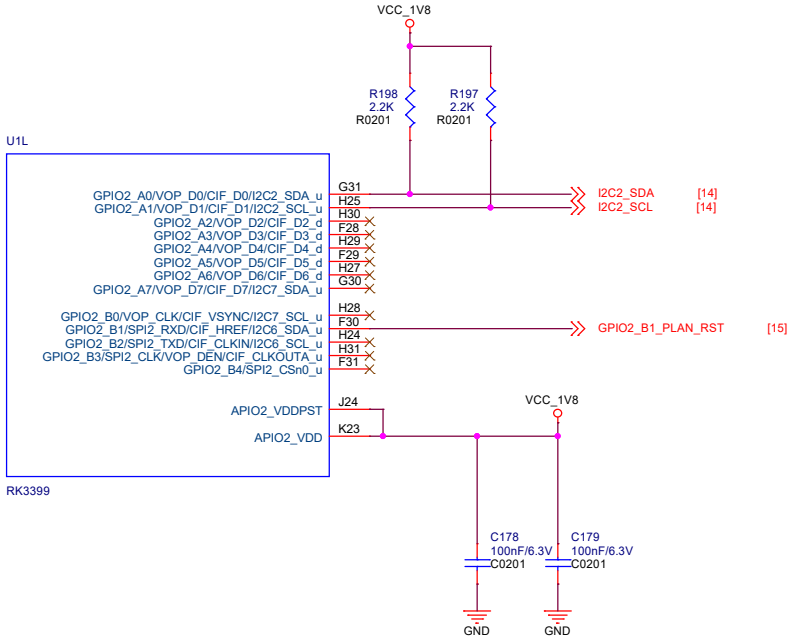
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

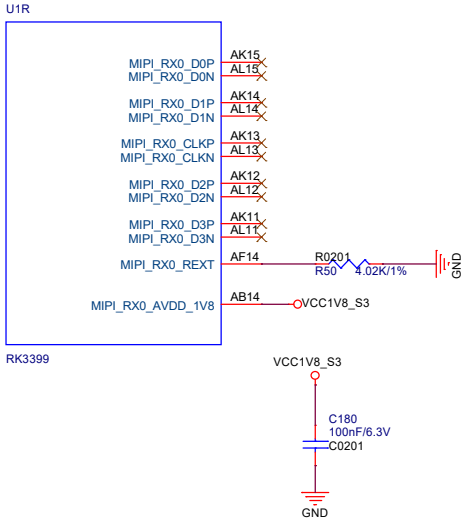
NanoPi R4SE	Rev 2205
Size A3	Page Name 08.RK3399 Display Interface
Date: Wednesday, May 25, 2022	Sheet: 8 / 21

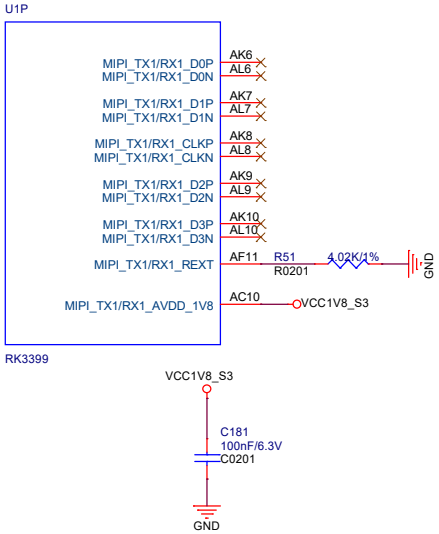


FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 09.RK3399 USB/USIC Controler
Date: Wednesday, May 25, 2022	Sheet: 9 / 21

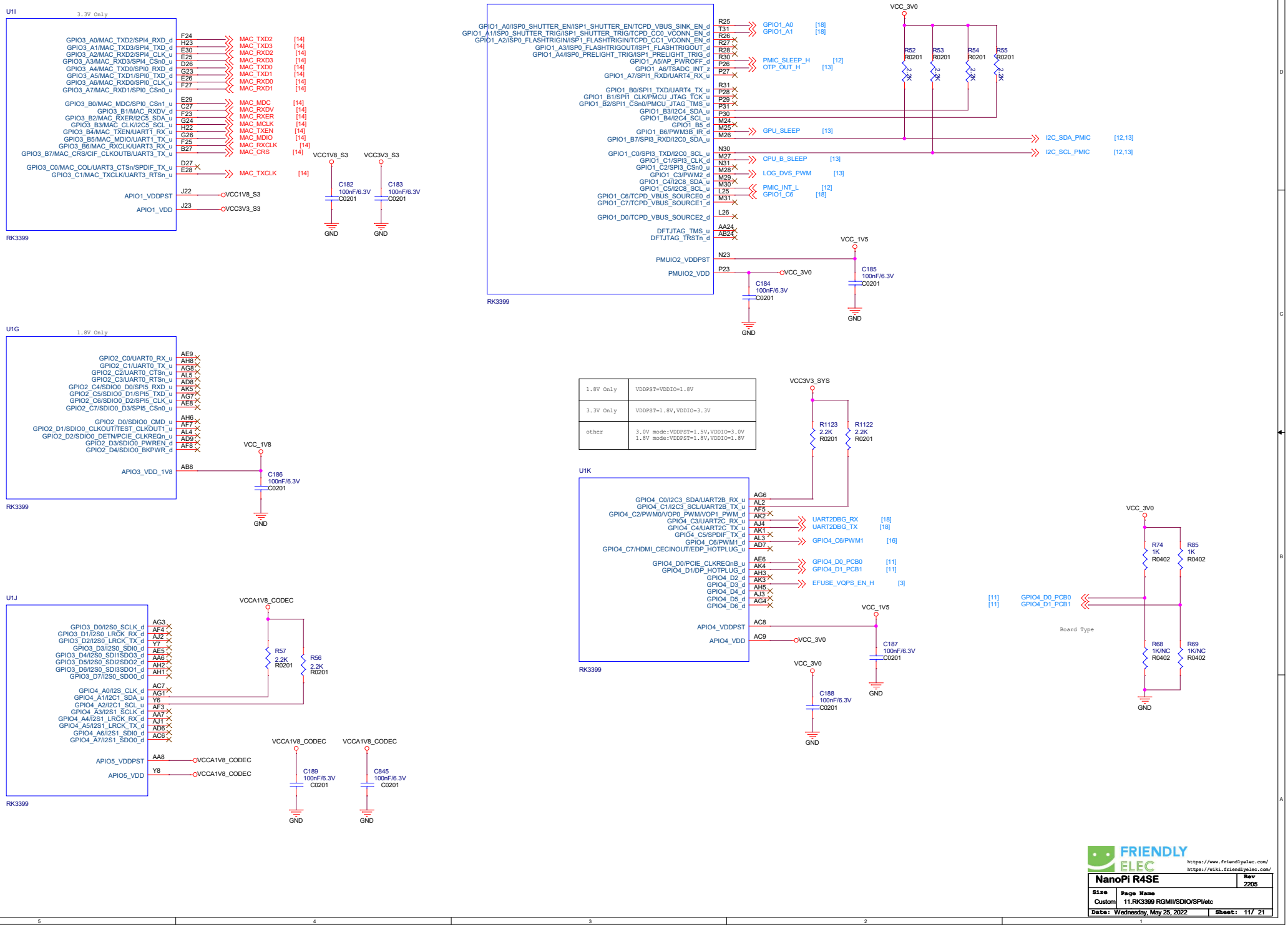


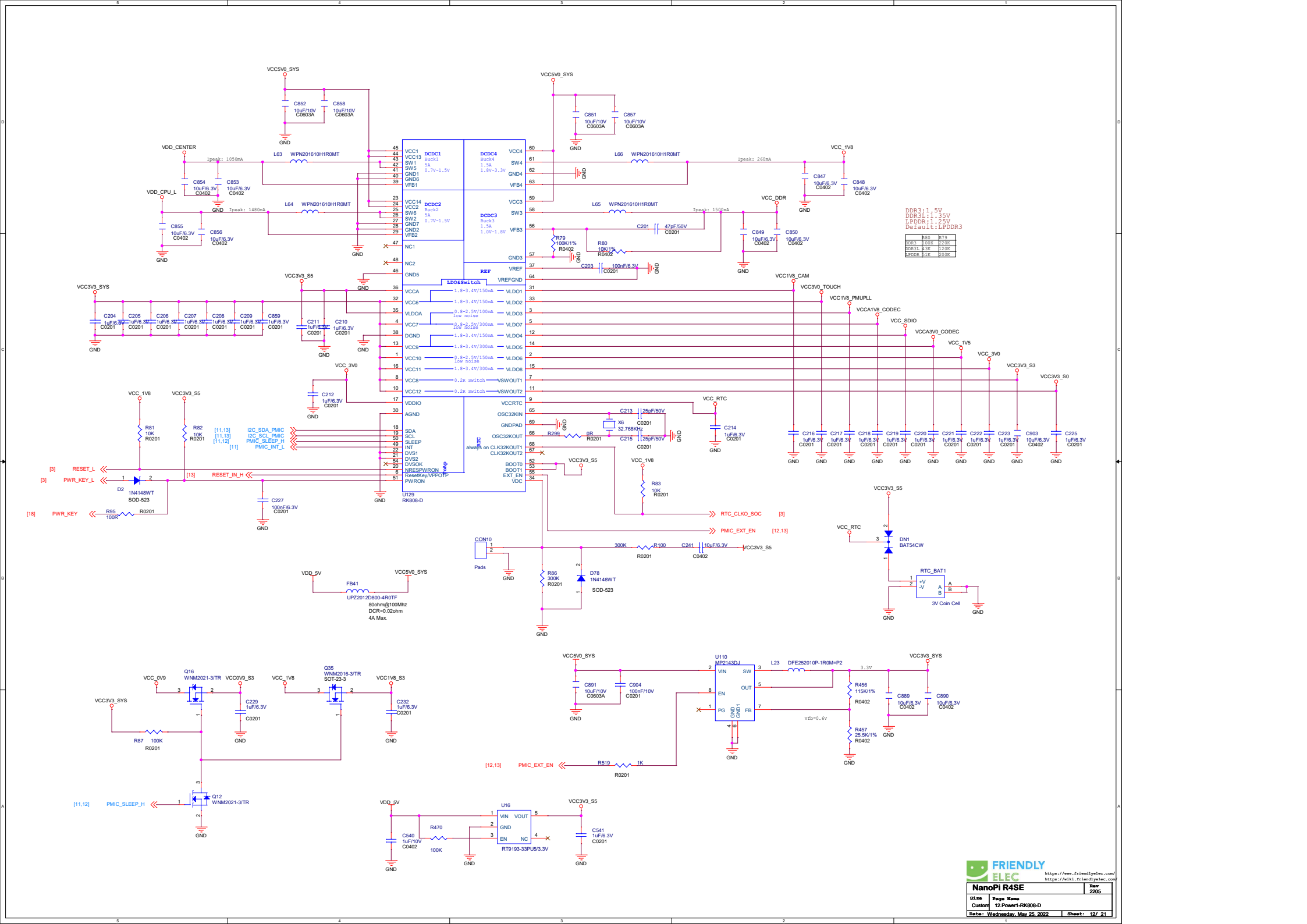


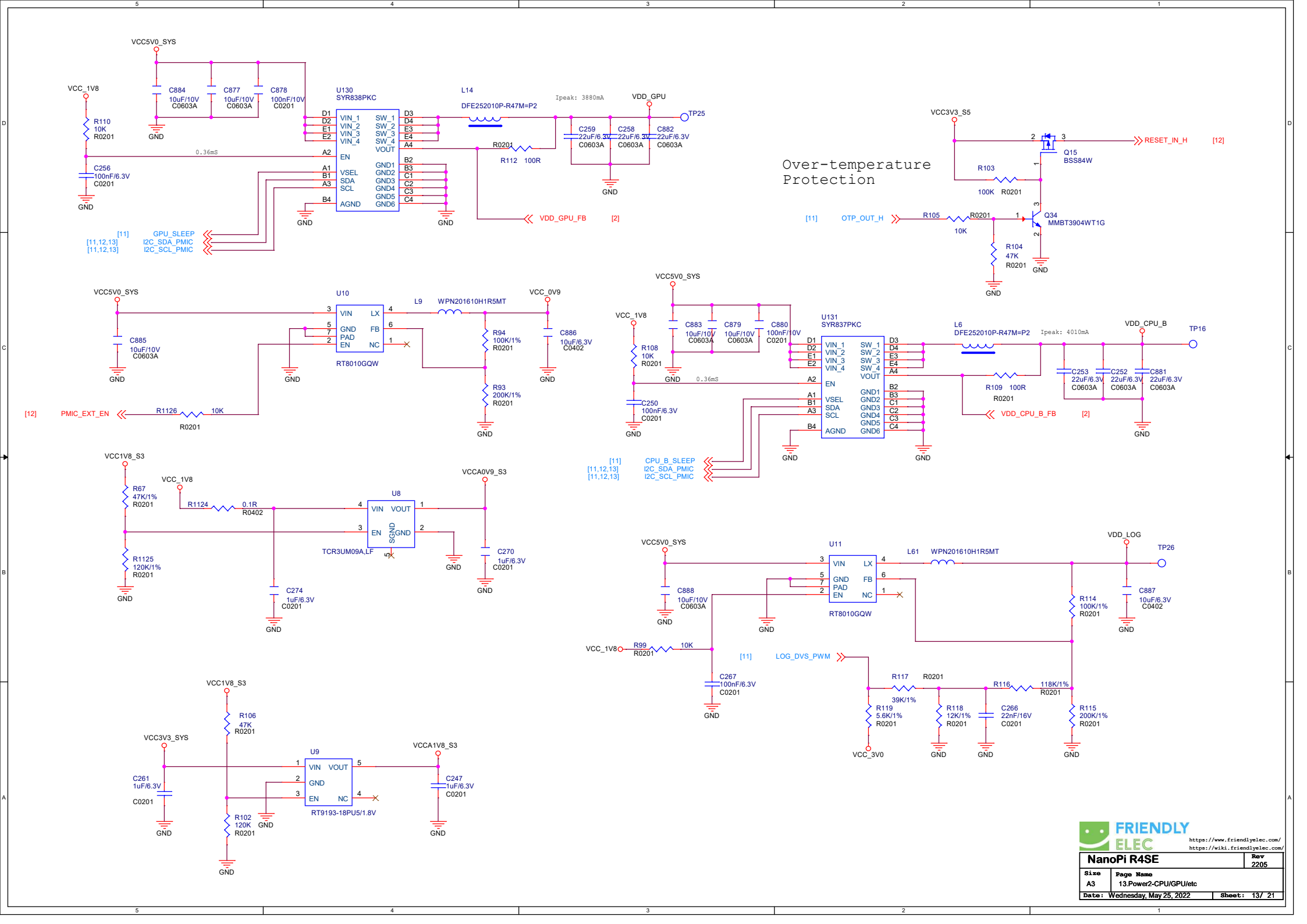


FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 10.RK3399 DVP/MIPI CSI
Date: Wednesday, May 25, 2022	Sheet: 10/ 21


U1I
3.3V Only
GPIO3_A0/MAC_TXD2/SPI4_RXD_d
GPIO3_A1/MAC_TXD3/SPI4_TXD_d
GPIO3_A2/MAC_RXD2/SPI4_CLK_u
GPIO3_A3/MAC_RXD3/SPI4_CSn0_u
GPIO3_A4/MAC_TXD0/SPI0_RXD_d
GPIO3_A5/MAC_TXD1/SPI0_TXD_d
GPIO3_A6/MAC_RXD0/SPI0_CLK_u
GPIO3_A7/MAC_RXD1/SPI0_CSn0_u
GPIO3_B0/MAC_MDC/SPI0_CSn1_u
GPIO3_B1/MAC_RXDV_d
GPIO3_B2/MAC_RXER/I2C5_SDA_u
GPIO3_B3/MAC_CLK/I2C5_SCL_u
GPIO3_B4/MAC_TXEN/UART1_RX_u
GPIO3_B5/MAC_MDIO/UART1_TX_u
GPIO3_B6/MAC_RXCLK/UART3_RX_u
GPIO3_B7/MAC_CRS/CIF_CLKOUTB/UART3_TX_u
GPIO3_C0/MAC_COL/UART3_CTSn/SPDIF_TX_u
GPIO3_C1/MAC_TXCLK/UART3_RTSn_u
APIO1_VDDPST
APIO1_VDD
F24 H23 E30 E25 D26 G23 E26 F27 E29 C27 F23 G24 H22 G26 F25 B27 D27 E28 J22 J23
MAC_TXD2 [14]
MAC_TXD3 [14]
MAC_RXD2 [14]
MAC_RXD3 [14]
MAC_TXD0 [14]
MAC_TXD1 [14]
MAC_RXD0 [14]
MAC_RXD1 [14]
MAC_MDC [14]
MAC_RXDV [14]
MAC_RXER [14]
MAC_MCLK [14]
MAC_TXEN [14]
MAC_MDIO [14]
MAC_RXCLK [14]
MAC_CRS [14]
MAC_TXCLK [14]
VCC1V8_S3
VCC3V3_S3
C182 100nF/6.3V C0201
C183 100nF/6.3V C0201
GND
RK3399
U1E
GPIO1_A0/ISP0_SHUTTER_EN/ISP1_SHUTTER_EN/TCPD_VBUS_SINK_EN_d
GPIO1_A1/ISP0_SHUTTER_TRIG/ISP1_SHUTTER_TRIG/TCPD_CC0_VCONN_EN_d
GPIO1_A2/ISP0_FLASHTRIGIN/ISP1_FLASHTRIGIN/TCPD_CC1_VCONN_EN_d
GPIO1_A3/ISP0_FLASHTRIGOUT/ISP1_FLASHTRIGOUT_d
GPIO1_A4/ISP0_PRELIGHT_TRIG/ISP1_PRELIGHT_TRIG_d
GPIO1_A5/AP_PWROFF_d
GPIO1_A6/TSADC_INT_z
GPIO1_A7/SPI1_RXD/UART4_RX_u
GPIO1_B0/SPI1_TXD/UART4_TX_u
GPIO1_B1/SPI1_CLK/PMCU_JTAG_TCK_u
GPIO1_B2/SPI1_CSn0/PMCU_JTAG_TMS_u
GPIO1_B3/I2C4_SDA_u
GPIO1_B4/I2C4_SCL_u
GPIO1_B5_d
GPIO1_B6/PWM3B_IR_d
GPIO1_B7/SPI3_RXD/I2C0_SDA_u
GPIO1_C0/SPI3_TXD/I2C0_SCL_u
GPIO1_C1/SPI3_CLK_d
GPIO1_C2/SPI3_CSn0_u
GPIO1_C3/PWM2_d
GPIO1_C4/I2C8_SDA_u
GPIO1_C5/I2C8_SCL_u
GPIO1_C6/TCPD_VBUS_SOURCE0_d
GPIO1_C7/TCPD_VBUS_SOURCE1_d
GPIO1_D0/TCPD_VBUS_SOURCE2_d
DFTJTAG_TMS_u
DFTJTAG_TRSTn_d
PMUIO2_VDDPST
PMUIO2_VDD
R25 T31 R26 R27 R28 R30 P26 P27 R31 P28 P29 P31 P30 M24 M25 M26 N30 M27 N31 M28 M29 M30 L25 M31 L26 AA24 AB24 N23 P23
GPIO1_A0 [18]
GPIO1_A1 [18]
PMIC_SLEEP_H [12]
OTP_OUT_H [13]
GPU_SLEEP [13]
CPU_B_SLEEP [13]
LOG_DVS_PWM [13]
PMIC_INT_L [12]
GPIO1_C6 [18]
I2C_SDA_PMIC [12,13]
I2C_SCL_PMIC [12,13]
VCC_3V0
R52 R53 R54 R55 R0201 2.2K
VCC_1V5
C184 100nF/6.3V C0201
C185 100nF/6.3V C0201
GND
RK3399
U1G
1.8V Only
GPIO2_C0/UART0_RX_u
GPIO2_C1/UART0_TX_u
GPIO2_C2/UART0_CTSn_u
GPIO2_C3/UART0_RTSn_u
GPIO2_C4/SDIO0_D0/SPI5_RXD_u
GPIO2_C5/SDIO0_D1/SPI5_TXD_u
GPIO2_C6/SDIO0_D2/SPI5_CLK_u
GPIO2_C7/SDIO0_D3/SPI5_CSn0_u
GPIO2_D0/SDIO0_CMD_u
GPIO2_D1/SDIO0_CLKOUT/TEST_CLKOUT1_u
GPIO2_D2/SDIO0_DETN/PCIE_CLKREQn_u
GPIO2_D3/SDIO0_PWREN_d
GPIO2_D4/SDIO0_BKPWR_d
APIO3_VDD_1V8
AE9 AH8 AG8 AL5 AD8 AK5 AG7 AE8 AH6 AF7 AL4 AD9 AF8 AB8
VCC_1V8
C186 100nF/6.3V C0201
GND
RK3399
1.8V Only | VDDPST=VDDIO=1.8V
3.3V Only | VDDPST=1.8V,VDDIO=3.3V
other | 3.0V mode:VDDPST=1.5V,VDDIO=3.0V 1.8V mode:VDDPST=1.8V,VDDIO=1.8V
VCC3V3_SYS
R1123 2.2K R0201
R1122 2.2K R0201
U1K
GPIO4_C0/I2C3_SDA/UART2B_RX_u
GPIO4_C1/I2C3_SCL/UART2B_TX_u
GPIO4_C2/PWM0/VOP0_PWM/VOP1_PWM_d
GPIO4_C3/UART2C_RX_u
GPIO4_C4/UART2C_TX_u
GPIO4_C5/SPDIF_TX_d
GPIO4_C6/PWM1_d
GPIO4_C7/HDMI_CECINOUT/EDP_HOTPLUG_u
GPIO4_D0/PCIE_CLKREQnB_u
GPIO4_D1/DP_HOTPLUG_d
GPIO4_D2_d
GPIO4_D3_d
GPIO4_D4_d
GPIO4_D5_d
GPIO4_D6_d
APIO4_VDDPST
APIO4_VDD
AG6 AL2 AF5 AK2 AJ4 AK1 AL3 AD7 AE6 AK4 AH3 AK3 AH5 AJ3 AG4 AC8 AC9
UART2DBG_RX [18]
UART2DBG_TX [18]
GPIO4_C6/PWM1 [16]
GPIO4_D0_PCB0 [11]
GPIO4_D1_PCB1 [11]
EFUSE_VQPS_EN_H [3]
VCC_1V5
C187 100nF/6.3V C0201
VCC_3V0
C188 100nF/6.3V C0201
GND
RK3399
VCC_3V0
R74 1K R0402
R85 1K R0402
R68 1K/NC R0402
R69 1K/NC R0402
[11] GPIO4_D0_PCB0
[11] GPIO4_D1_PCB1
Board Type
GND
U1J
GPIO3_D0/I2S0_SCLK_d
GPIO3_D1/I2S0_LRCK_RX_d
GPIO3_D2/I2S0_LRCK_TX_d
GPIO3_D3/I2S0_SDI0_d
GPIO3_D4/I2S0_SDI1SDO3_d
GPIO3_D5/I2S0_SDI2SDO2_d
GPIO3_D6/I2S0_SDI3SDO1_d
GPIO3_D7/I2S0_SDO0_d
GPIO4_A0/I2S_CLK_d
GPIO4_A1/I2C1_SDA_u
GPIO4_A2/I2C1_SCL_u
GPIO4_A3/I2S1_SCLK_d
GPIO4_A4/I2S1_LRCK_RX_d
GPIO4_A5/I2S1_LRCK_TX_d
GPIO4_A6/I2S1_SDI0_d
GPIO4_A7/I2S1_SDO0_d
APIO5_VDDPST
APIO5_VDD
AG3 AF4 AJ2 Y7 AE5 AA6 AH2 AH1 AC7 AG1 Y6 AF3 AA7 AJ1 AD6 AC6 AA8 Y8
VCCA1V8_CODEC
R57 2.2K R0201
R56 2.2K R0201
C189 100nF/6.3V C0201
C845 100nF/6.3V C0201
GND
RK3399
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/
NanoPi R4SE
Rev 2205
Size Custom
Page Name 11.RK3399 RGMII/SDIO/SPI/etc
Date: Wednesday, May 25, 2022
Sheet: 11/ 21

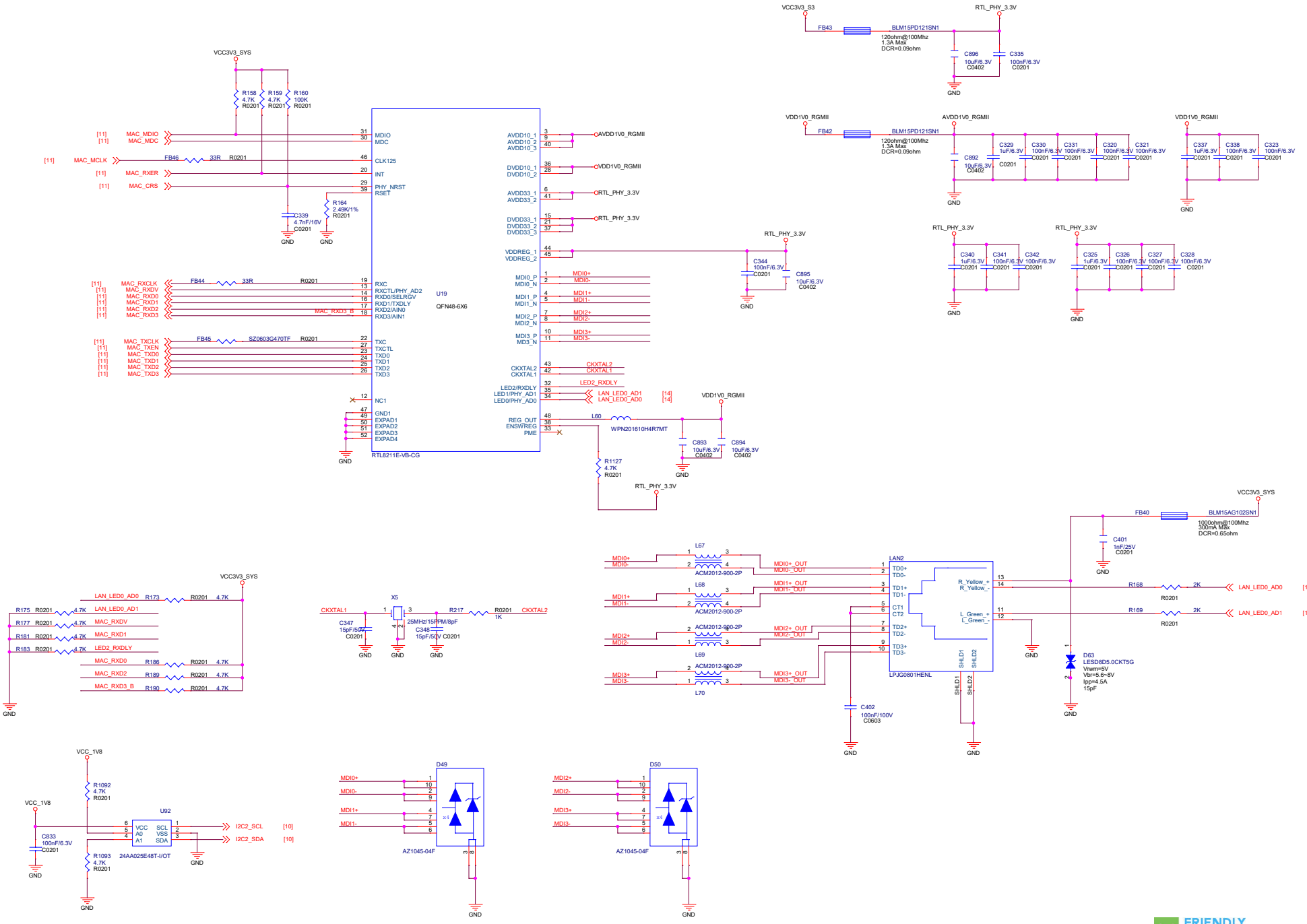

VCC5V0_SYS
C852 10uF/10V C0603A
C858 10uF/10V C0603A
GND
VDD_CENTER
Ipeak: 1050mA
L63 WPN201610H1R0MT
C854 10uF/6.3V C0402
C853 10uF/6.3V C0402
VDD_CPU_L
Ipeak: 1480mA
L64 WPN201610H1R0MT
C855 10uF/6.3V C0402
C856 10uF/6.3V C0402
VCC1 VCC13 SW1 SW5 GND1 GND6 VFB1
DCDC1 Buck1 5A 0.7V-1.5V
VCC14 VCC2 SW6 SW2 GND7 GND2 VFB2
DCDC2 Buck2 5A 0.7V-1.5V
NC1
NC2
GND5
VCCA VCC6 VLDOA VCC7 DGND VCC9 VCC10 VCC11 VCC8 VCC12
LDO&Switch
VDDIO AGND
SDA SCL SLEEP INT DVS1 DVS2 DVSOK NRESPWRON ResetKey/VPPOTP PWRON
U129 RK808-D
VCC3V3_SYS
C204 1uF/6.3V C0201
C205 1uF/6.3V C0201
C206 1uF/6.3V C0201
C207 1uF/6.3V C0201
C208 1uF/6.3V C0201
C209 1uF/6.3V C0201
C859 1uF/6.3V C0201
VCC3V3_S5
C211 1uF/6.3V C0201
C210 1uF/6.3V C0201
VCC_3V0
C212 1uF/6.3V C0201
VCC_1V8
VCC3V3_S5
R81 10K R0201
R82 10K R0201
[11,13] I2C_SDA_PMIC
[11,13] I2C_SCL_PMIC
[11,12] PMIC_SLEEP_H
[11] PMIC_INT_L
[3] RESET_L
[3] PWR_KEY_L
[13] RESET_IN_H
D2 1N4148WT SOD-523
[18] PWR_KEY
R95 100R
R0201
C227 100nF/6.3V C0201
VCC5V0_SYS
C851 10uF/10V C0603A
C857 10uF/10V C0603A
DCDC4 Buck4 1.5A 1.8V-3.3V
VCC4 SW4 GND4 VFB4
L66 WPN201610H1R0MT
Ipeak: 260mA
VCC_1V8
C847 10uF/6.3V C0402
C848 10uF/6.3V C0402
DCDC3 Buck3 1.5A 1.0V-1.8V
VCC3 SW3 VFB3
L65 WPN201610H1R0MT
Ipeak: 1500mA
VCC_DDR
C849 10uF/6.3V C0402
C850 10uF/6.3V C0402
R79 100K/1% R0402
R80 10K/1% R0402
C201 47pF/50V C0201
C203 100nF/6.3V C0201
GND3
REF
VREF
VREFGND
DDR3:1.5V
DDR3L:1.35V
LPDDR:1.25V
Default:LPDDR3
VLDO1 1.8-3.4V/150mA
VLDO2 1.8-3.4V/150mA
VLDO3 0.8-2.5V/100mA low noise
VLDO7 0.8-2.5V/300mA low noise
VLDO4 1.8-3.4V/150mA
VLDO5 1.8-3.4V/300mA
VLDO6 0.8-2.5V/150mA low noise
VLDO8 1.8-3.4V/300mA
VSWOUT1 0.2R Switch
VSWOUT2 0.2R Switch
VCCRTC
OSC32KIN
GNDPAD
OSC32KOUT
always on CLK32KOUT1
CLK32KOUT2
BOOT0
BOOT1
EXT_EN
VDC
C213 25pF/50V C0201
X6 32.768KHz
C215 25pF/50V C0201
R299 0R R0201
VCC_RTC
C214 1uF/6.3V C0201
R83 10K R0201
RTC_CLKO_SOC [3]
PMIC_EXT_EN [12,13]
VCC1V8_CAM
VCC3V0_TOUCH
VCC1V8_PMUPLL
VCCA1V8_CODEC
VCC_SDIO
VCCA3V0_CODEC
VCC_1V5
VCC_3V0
VCC3V3_S3
VCC3V3_S0
C216 1uF/6.3V C0201
C217 1uF/6.3V C0201
C218 1uF/6.3V C0201
C219 1uF/6.3V C0201
C220 1uF/6.3V C0201
C221 1uF/6.3V C0201
C222 1uF/6.3V C0201
C223 1uF/6.3V C0201
C903 10uF/6.3V C0402
C225 1uF/6.3V C0201
CON10 Pads
300K R100 R0201
C241 10uF/6.3V C0402
R86 300K R0201
D78 1N4148WT SOD-523
VDD_5V
FB41 UPZ2012D800-4R0TF
80ohm@100Mhz
DCR=0.02ohm
4A Max
VCC5V0_SYS
DN1 BAT54CW
RTC_BAT1
3V Coin Cell
Q16 WNM2021-3/TR
VCC_0V9
VCC0V9_S3
C229 1uF/6.3V C0201
Q35 WNM2016-3/TR SOT-23-3
VCC_1V8
VCC1V8_S3
C232 1uF/6.3V C0201
R87 100K R0201
Q12 WNM2021-3/TR
[11,12] PMIC_SLEEP_H
U110 MP2143DJ
VIN SW OUT EN FB PG GND GND1
L23 DFE252010P-1R0M=P2
3.3V
R456 115K/1% R0402
R457 25.5K/1% R0402
Vfb=0.6V
C891 10uF/10V C0603A
C904 100nF/10V C0201
C889 10uF/6.3V C0402
C890 10uF/6.3V C0402
[12,13] PMIC_EXT_EN
R519 1K R0201
U16 RT9193-33PU5/3.3V
VIN VOUT GND EN NC
R470 100K
C540 1uF/10V C0402
C541 1uF/6.3V C0201
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/
NanoPi R4SE
Rev 2205
Size Custom
Page Name 12.Power1-RK808-D
Date: Wednesday, May 25, 2022
Sheet: 12/ 21



FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

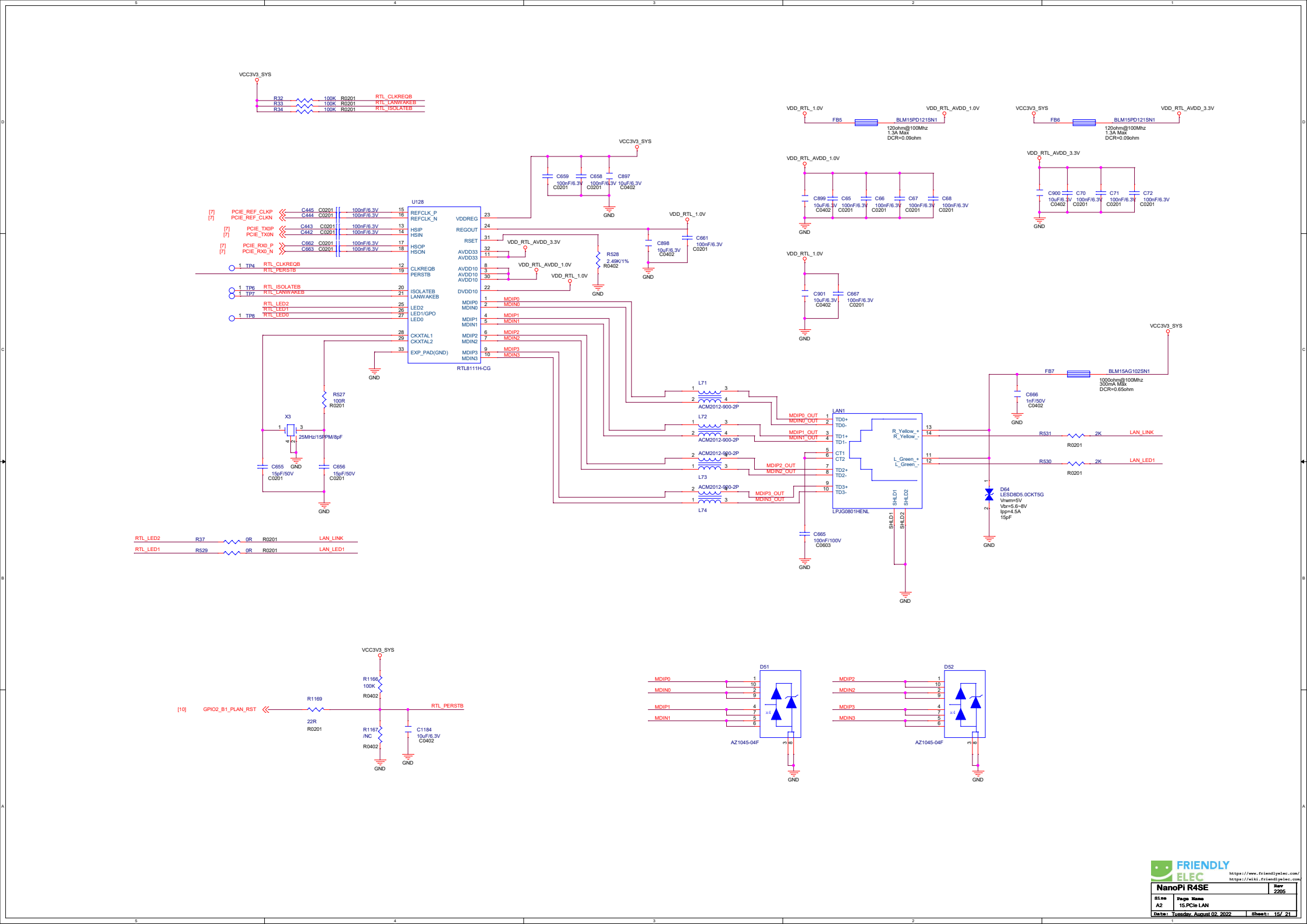
NanoPi R4SE	Rev 2205
Size A3	Page Name 13.Power2-CPU/GPU/etc
Date: Wednesday, May 25, 2022	Sheet: 13/ 21

10/100/1000M Ethernet



FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

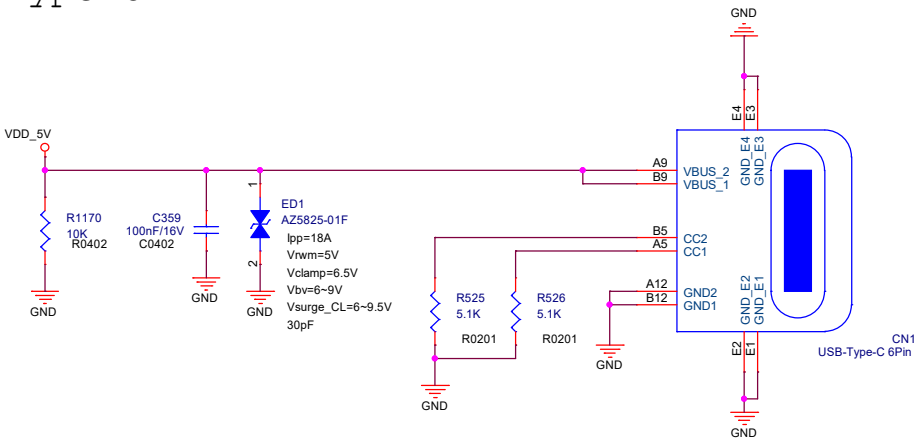
NanoPi R4SE		Rev 2205
Size A2	Page Name 14.RGMII-10/100/1000M	
Date: Wednesday, May 25, 2022		Sheet: 14/ 21



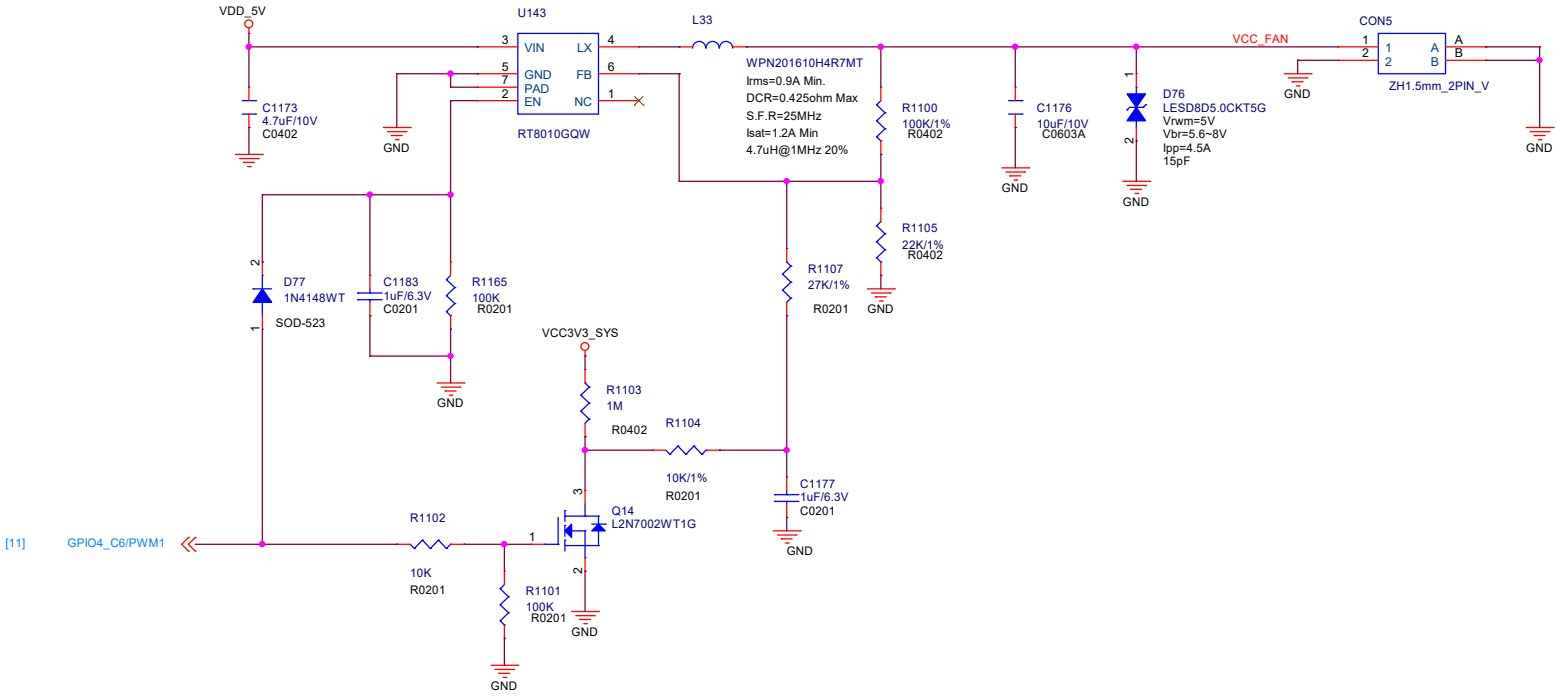
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com

NanoPi R4SE		Rev 2205
Size A2	Page Name 15.PCIe LAN	
Date: Tuesday, August 02, 2022		Sheet: 15/ 21

USB Type-C

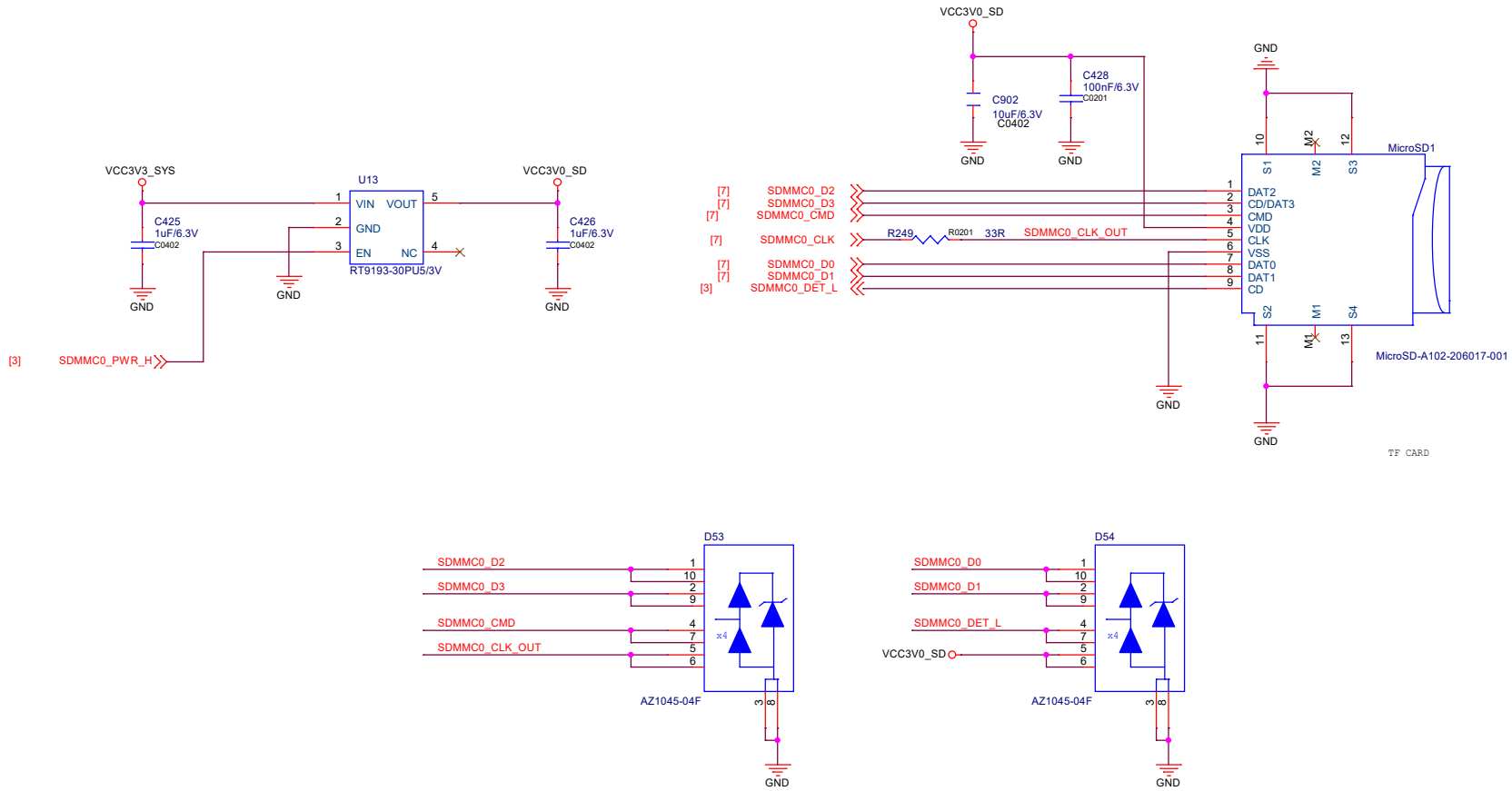


FAN



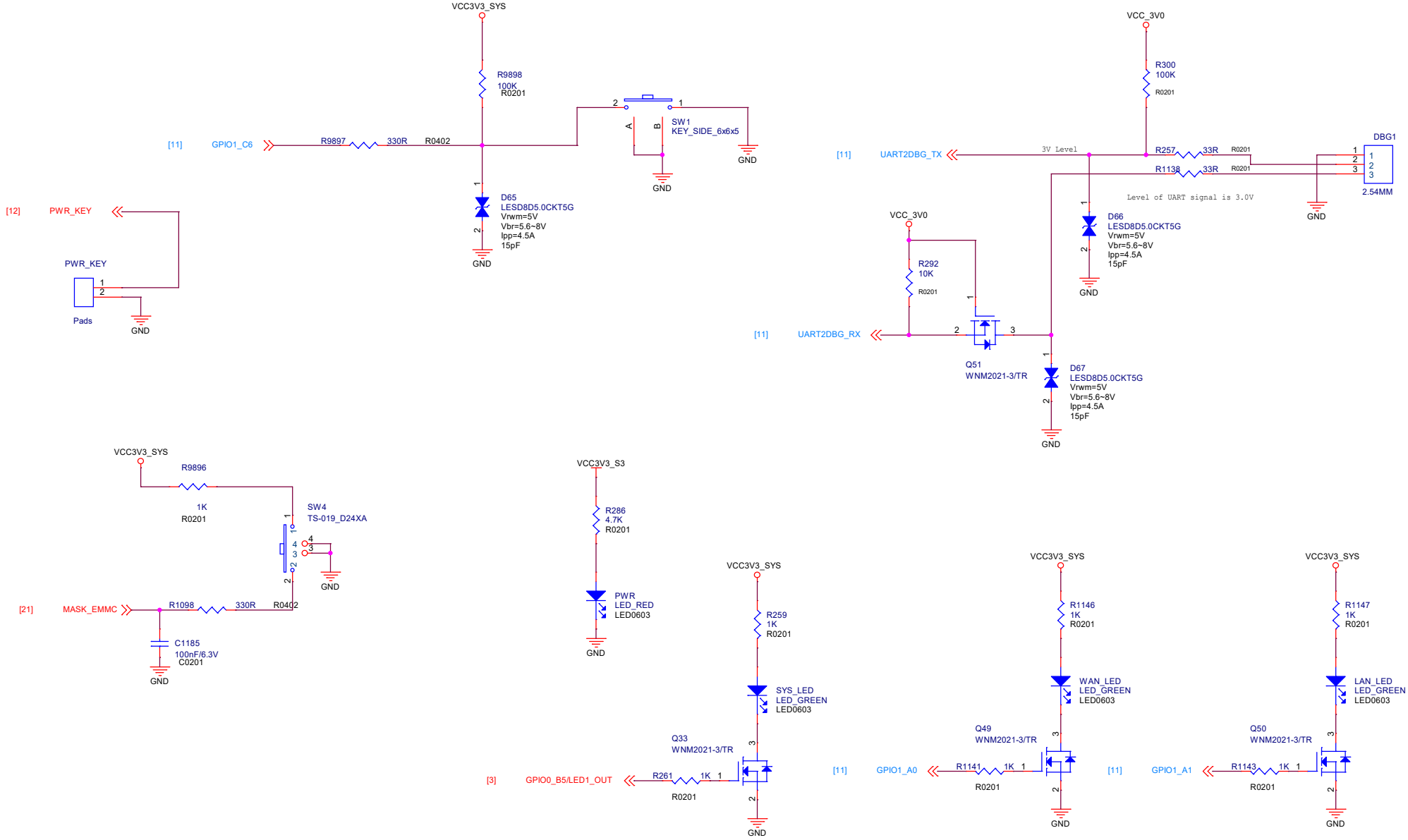
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE		Rev 2205
Size A3	Page Name 16.USB Type-C/FAN/GPIO	
Date: Wednesday, May 25, 2022		Sheet: 16/ 21



FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE		Rev 2205
Size A3	Page Name 17.SD Card	
Date: Wednesday, May 25, 2022		Sheet: 17/ 21



FRIENDLY ELEC

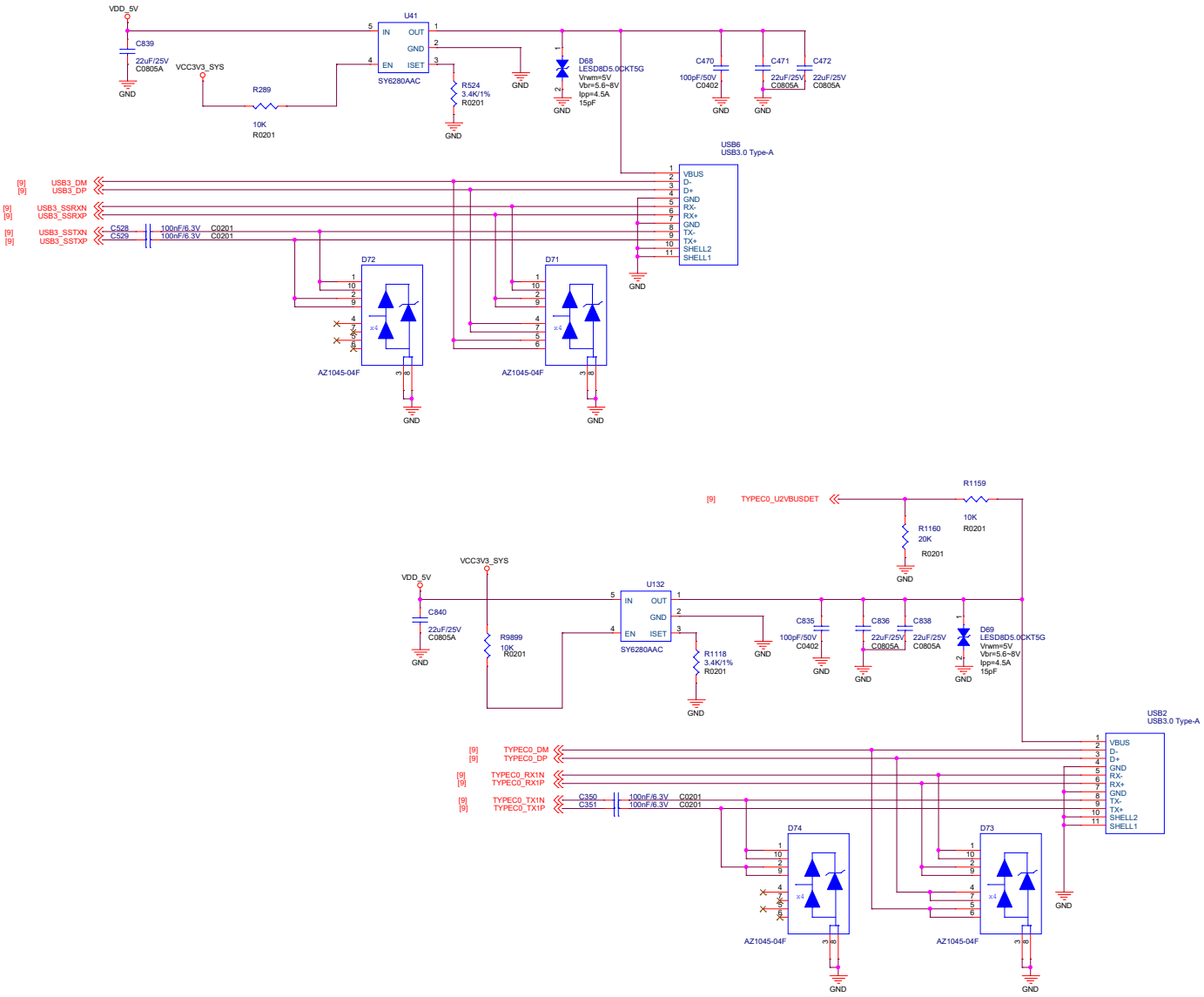
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 18.Key/LED/Debug
Date: Wednesday, May 25, 2022	Sheet: 18/ 21

5 4 3 2 1

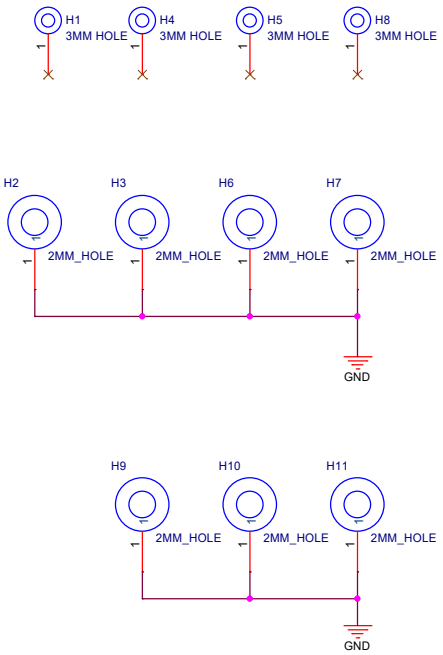
D C B A

USB3.0 Host



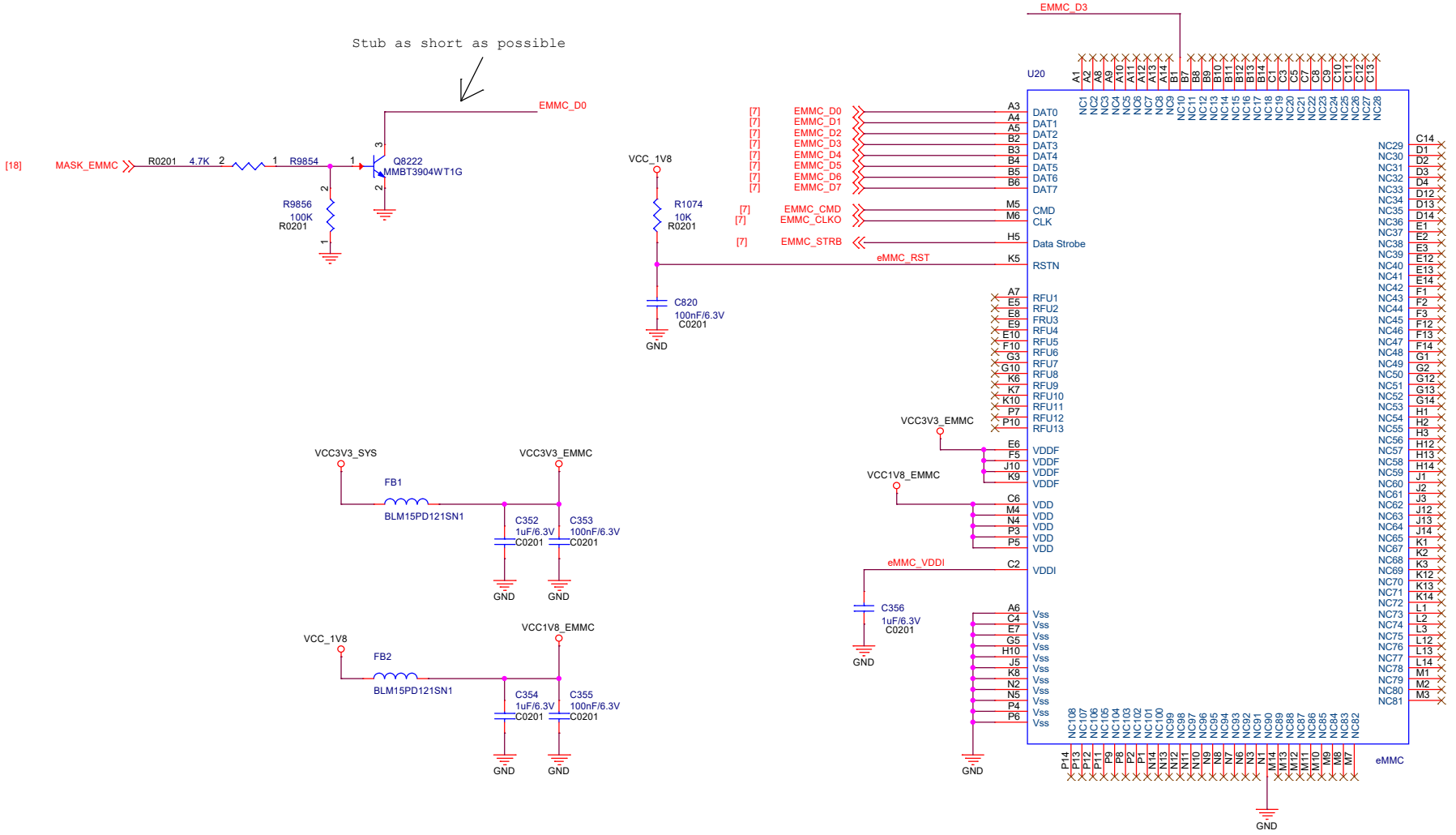
FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE		Rev 2205
Size A2	Page Name 19.USB Ports	
Date: Wednesday, May 25, 2022	Sheet: 19/ 21	



FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 20.Holes
Date: Wednesday, May 25, 2022	Sheet: 20/ 21



FRIENDLY ELEC
https://www.friendlyelec.com/
https://wiki.friendlyelec.com/

NanoPi R4SE	Rev 2205
Size A3	Page Name 21.eMMC
Date: Tuesday, July 12, 2022	Sheet: 21/ 21